MGU-UGP BSc (HONS) MICROBIOLOGY

III SEMESTER

DSE- MG3DSEMBG201– EXTREMOPHILES,GEO & ASTROMICROBIOLOGY

(2024 ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall attainment of the outcomes.

Time: 2 Hours **Max. marks: 70**

Part A
Very Short Answer Questions
Answer all questions
Each question carries 1 mark.

1. Define extremophiles. [K][1]
2. Write down permafrost. [U][1]
3. Explain bioleaching. [U][2]
4. Explain onemedical application of extremophiles. [A][4]
5. Point outany one effect of space on microbes.[An][3] (1x5=5)

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks.

6. Distinguish thermophiles and psychrophiles. [U][1]
7. Give two examples of microbes that solubilise phosphate. [U][2]
8. What are methanogens?[K][1]
9. Write down any two ways that microorganisms contribute to soil formation. [U][2]
10. Distinguish bioleaching and biomining. [U][2]
11. Write down any two adaptations of microorganisms in the polar environment. [U][1]
12. List the main goal of the Chandrayan mission. [An][3]
13. Explain the role of *Acidothiobacillusferroxidans* in bioleaching.[U][2]
14. Point out any two challenges in detecting life beyond Earth. [An][3]
15. Explain the advantages of using extremophiles in bioremediation.[A][4]
16. Write down any two enzymes produced by extremophiles. [U][4]
17. How do acidophiles help in detoxifying heavy metals? [U][4] (2x10=20)

Part C
Short Essay Questions
Answer any 5 questions
Each question carries 6 marks

18. Describe biocorrosion and give examples of microbes causing it. [K][2]
19. Explain the role of extremophiles in the carbon cycle. [U][2]

20. Classify extremophiles based on the environment they inhabit. [U][1]
21. Explain exomicrobiology and its relevance in space missions.[A][3]
22. List a brief account of instruments for life detection in space.[An][3]
23. Explain the role of extremophiles in biopharmaceuticals. [A][4]
24. Write down the structure of the cell wall of methanogens. [U][1]

(6x5=30)

Part D

Long Essay Questions

Answer any 1 question

Each question carries 15 marks

25. Summarisethe adaptations of extremophiles that allow them to survive in extreme conditions. [U][1]
26. Explain the role of extremophiles inthe Sulphur cyclewith the help of a diagram. [U][4]

(15x1=15)